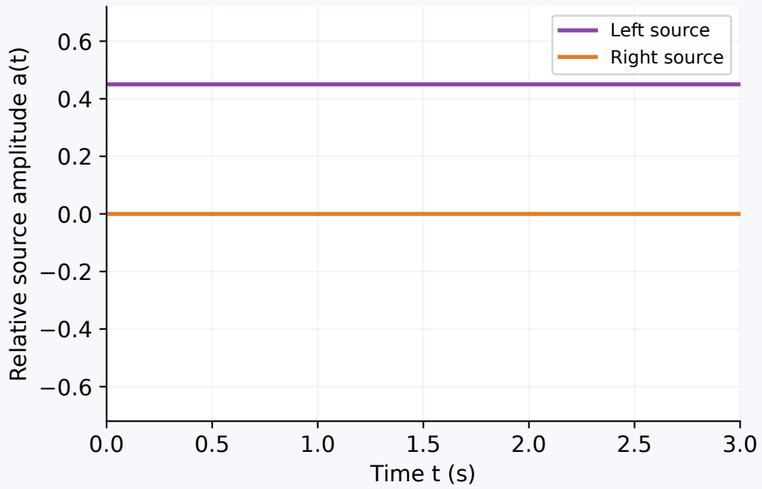
MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

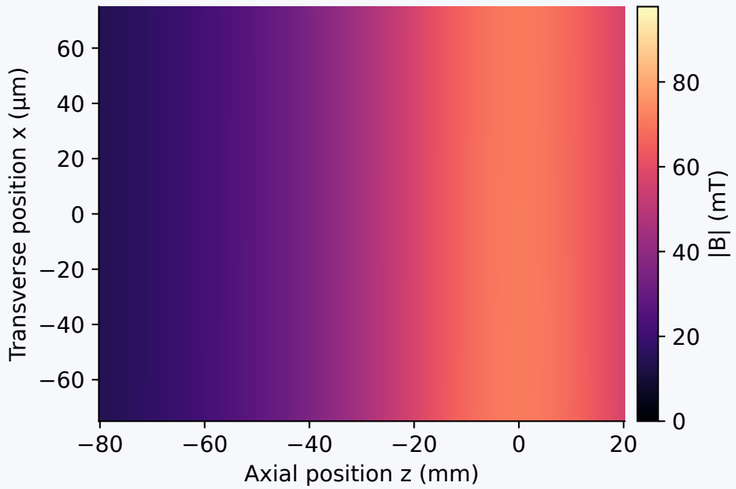
Constant left source

$a_L = 0.45, \quad a_R = 0$

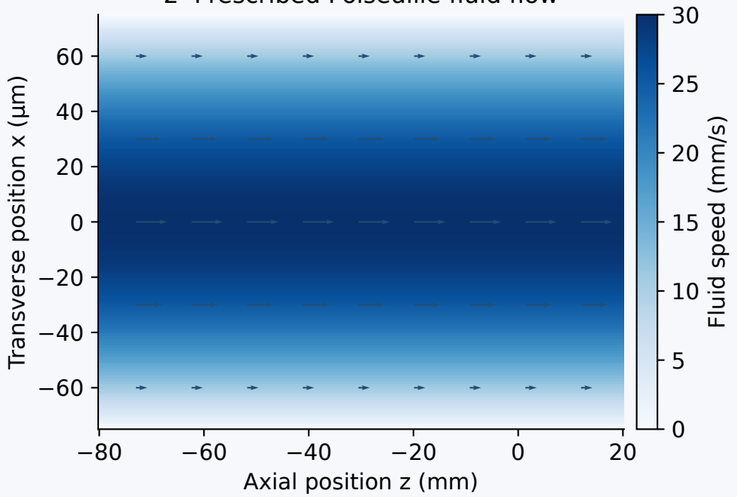
1a Prescribed magnetic amplitudes



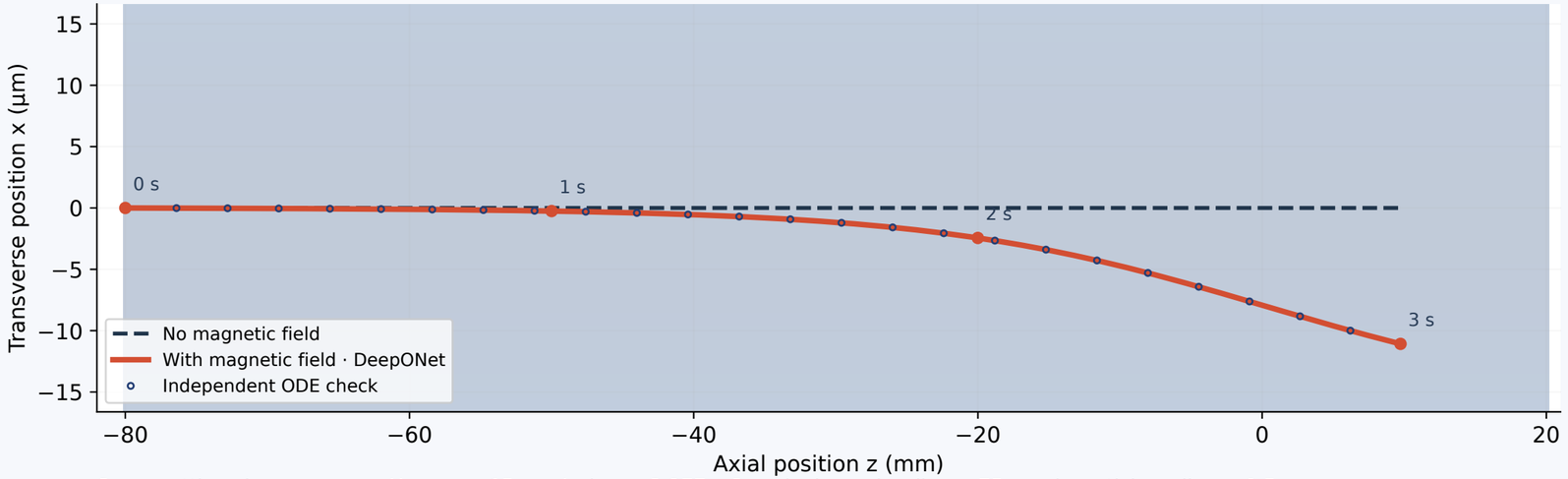
1b Magnetic field at t = 0.00 s



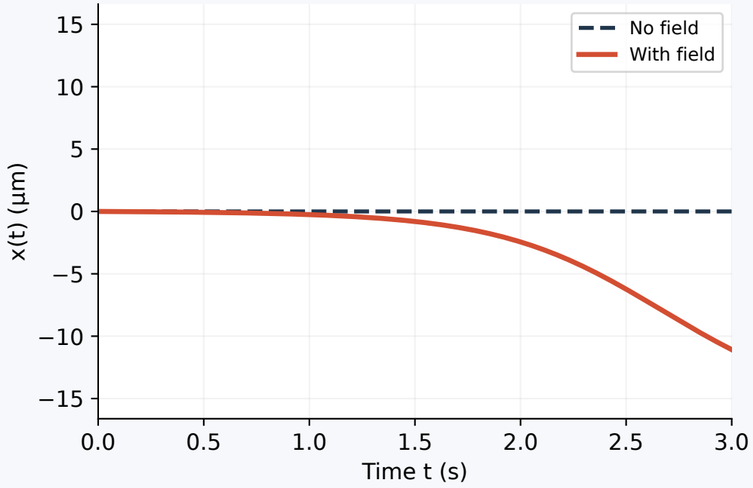
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 µm | particle radius = 0.5 µm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

Independent max error: x 0.001 µm · z 0.94 µm